

Avalon-MM SDRAM Controller IP Core

User Guide

IP core	<code>altera_avalon_mm_sdram_controller</code>
Core version	1.0
Document version	1.0
Date	August 2026
Target	Intel FPGA / Quartus Prime, Platform Designer

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1. About this core

1.1 Features

- **One open row per bank.** Four rows held open at once rather than one, so an access that changes bank costs no row command.
- **Read/write turnaround without a row cycle.** 0 cycles write → read, CAS+1 read → write, as the device datasheet describes.
- **Look-ahead row activation.** The row for the *next* buffered access is opened or closed while the current one is still being served.
- **Postponed refresh.** Up to `REF_MAX_PEND` refreshes are deferred to a quiet moment, then forced through.
- **Parameterised in time, not cycles.** Device timings are picoseconds; cycles are derived inside the HDL.
- **Drop-in.** Same `s1` slave, same `wire` conduit, same default address map as the core it replaces.

1.2 What it is for

Mixed and scattered traffic — which is what a CPU generates. Sequential streaming was already at 97% of the bus limit before this core existed and is not where the headroom was. See section 7.

1.3 Status

Simulation only. The core has never been synthesised or run on hardware: there is no Quartus licence in its development environment. There are therefore no resource figures and no `f_MAX` in this document, and none should be inferred. What has been measured is described in sections 7 and 8, and is reproducible.

2. Getting started

2.1 Adding the component

Add the repository to the IP search path — **Tools ► Options ► IP Search Path**, or `--search-path` on the command line — and the component appears in the IP Catalog under **Memory Interfaces and Controllers / Custom** as *Avalon-MM SDRAM Controller (per-bank rows)*.

2.2 Start from a preset

Select the preset for your part rather than typing timings in:

```
add_instance sdram altera_avalon_mm_sdram_controller
apply_preset sdram "ISSI IS42S16320D-7 - DE10-Lite 64 MByte"
```

A preset carries the geometry, the timings and the refresh figures for one device, taken from its datasheet once. Only parts whose timing has been checked against a datasheet *and* exercised through `benchmark/` are supplied; adding one is a block of XML in `altera_avalon_mm_sdram_controller.qprs`.

Caution

There is deliberately no "generic SDRAM" preset. A timing constant that is one speed grade optimistic produces a design that passes every simulation, works on the bench, and corrupts data once the board is warm.

2.3 Connecting it

Interface	Connect to
<code>clk</code> , <code>reset</code>	Your clock source. The clock rate is read from it , so no frequency is typed in anywhere
<code>s1</code>	The master, or the interconnect
<code>wire</code>	Export to the top level, and constrain the SDRAM pins

2.4 What is checked at generation time

The component refuses, or warns about, four configurations that would otherwise build and be wrong on hardware:

Condition	Result
<code>SA_BITS</code> too small for <code>ROW_BITS</code> , or below 11	Error
<code>tRAS</code> longer than <code>tRC</code> , or <code>tRRD</code> longer than <code>tRC</code>	Error
A refresh interval the controller cannot sustain	Error
<code>ADDR_MAP</code> set to conventional	Warning — every address moves

It also reports the memory size and the exact cycle counts the HDL will derive at your clock, so they can be read against the datasheet:

Info: Memory: 64 MByte - 4 banks x 8192 rows x 1024 columns x 16 bits.

Info: At 143.000 MHz the HDL will use: tRC=9 tRAS=6 tRP=3 tRCD=3 tRRD=3
tWR=3 tRFC=9 cycles, CAS=3, one refresh every 1118 cycles.

3. Functional description

3.1 Initialisation

After reset the controller holds `za_waitrequest` high and runs the JEDEC power-up sequence: wait `T_INIT_US`, PRECHARGE ALL, `INIT_REFS` AUTO REFRESH commands, then LOAD MODE REGISTER. Only then is a transfer accepted.

The mode register is written with burst length 1, sequential burst type, and the configured CAS latency.

3.2 Serving a transfer

One command is issued per cycle, chosen in priority order — see Figure 3 in the block-diagram document. In summary: serve the head if its row is open, close the row if it is the wrong one, open the bank if it is closed, otherwise prepare the next access.

3.3 Read/write turnaround

A READ issued at cycle T has the device driving DQ at $T+CAS$. A WRITE drives DQ in its own cycle, so the earliest safe write is $T+CAS+1$. The other direction is free — the datasheet permits write data to be immediately followed by a READ command.

This is why mixed traffic settles at about 79 MB/s rather than 194: a write/read pair costs five cycles at CAS 3, and two accesses per five cycles is 80 MB/s. That is the device, not the scheduler.

3.4 Refresh

A timer accumulates outstanding refreshes at the interval implied by `REF_ROWS` and `REF_PERIOD_MS`. They are issued when the command buffer is empty, or forced once `REF_MAX_PEND` have accumulated. A refresh precharges every bank, so all open rows are lost and reopened as needed.

3.5 Byte enables

`az_be_n` is active low and is driven onto `zs_dqm` for writes. DQM is held low at all other times, which satisfies the SDR read-output enable requirement.

4. Parameters

4.1 Geometry

Parameter	Type	Default	Range	Description
DATA_BITS	Integer	16	8, 16, 32	SDRAM data bus width. One DQM bit per byte lane
ROW_BITS	Integer	13	10:15	Row address width
COL_BITS	Integer	10	8:11	Column address width. Column bit 10 steps over A10
BANK_BITS	Integer	2	1, 2, 3	Bank address width. Also how many rows can be open at once
SA_BITS	Integer	13	11:15	Address pins on the device. At least ROW_BITS , and at least 11
ADDR_W	Integer	25	derived	Avalon word address width — row + column + bank. Do not override

4.2 Device timing

Given in **picoseconds** in the HDL and in **nanoseconds** in Platform Designer, which scales them. See section 4.5 for why.

Parameter	Default	Description
T_RC_PS	60 000	ACTIVATE to ACTIVATE, same bank
T_RAS_PS	37 000	ACTIVATE to PRECHARGE, same bank
T_RP_PS	15 000	PRECHARGE to ACTIVATE, same bank
T_RCD_PS	15 000	ACTIVATE to READ or WRITE
T_RRD_PS	14 000	ACTIVATE to ACTIVATE, different bank
T_WR_PS	14 000	Write recovery: last write data to PRECHARGE
T_MRD_PS	14 000	LOAD MODE REGISTER to next command
T_RFC_PS	60 000	AUTO REFRESH to next ACTIVATE or REFRESH
CAS_LAT	3	CAS latency, 2 or 3

4.3 Initialisation and refresh

Parameter	Default	Range	Description
T_INIT_US	100	1:10000	Power-up wait before the first command, microseconds
INIT_REFS	8	2:15	AUTO REFRESH commands during initialisation
REF_ROWS	8192	512:32768	Rows the device refreshes per period
REF_PERIOD_MS	64	1:1000	Refresh period, milliseconds
REF_MAX_PEND	8	1:8	Refreshes that may be postponed. 1 refreshes immediately

4.4 Controller options

Parameter	Default	Range	Description
CLK_KHZ	100 000	derived	Taken from the connected clock source. Every timing is converted against it
ADDR_MAP	0	0, 1	0 = compatible with the Intel core, 1 = conventional {row, bank, column}
FIFO_DEPTH	8	2, 4, 8, 16, 32	Buffered commands. Also what look-ahead looks into
LOOKAHEAD	1	0, 1	Open or close the next access's row early
RD_EXTRA_LAT	0	0:3	Extra read-capture delay. Zero is correct for a direct connection

4.5 Why picoseconds, and why not cycles

Cycle counts are deliberately not exposed. A timing parameter rounded the wrong way is silent data corruption at temperature months later, not a clean failure, and pushing that arithmetic onto every integrator guarantees someone gets it wrong.

Picoseconds rather than a `real` number of nanoseconds because **Platform Designer emits a FLOAT parameter into the generated Verilog as a quoted string** — `.T_RC_NS("60.0")`. Assigned to a `parameter real`, that string is its ASCII bytes read as a number: 60.0 arrives as 909127216.0, and a 6-cycle tRC becomes 90 million. Integers cross the tool boundary intact.

Caution

If you add a parameter to this core, check the generated wrapper and confirm it arrives unquoted.

5. Signals

20 ports.

5.1 Clock and reset

Signal	Direction	Width	Description
clk	Input	1	All logic is in this domain
reset_n	Input	1	Asynchronous assert, synchronous deassert

5.2 `s1` — Avalon-MM slave

Word-addressed, `isMemoryDevice`, variable read latency with `readdatavalid`.

Signal	Direction	Width	Avalon role
az_addr	Input	ADDR_W	address (words)
az_be_n	Input	DATA_BITS/8	byteenable_n
az_cs	Input	1	chipselct
az_data	Input	DATA_BITS	writedata
az_rd_n	Input	1	read_n
az_wr_n	Input	1	write_n
za_data	Output	DATA_BITS	readdata
za_valid	Output	1	readdatavalid
za_waitrequest	Output	1	waitrequest

5.3 `wire` — SDRAM conduit

Signal	Direction	Width
zs_addr	Output	SA_BITS
zs_ba	Output	BANK_BITS
zs_cas_n	Output	1
zs_cke	Output	1
zs_cs_n	Output	1
zs_dq	Bidir	DATA_BITS
zs_dqm	Output	DATA_BITS/8
zs_ras_n	Output	1
zs_we_n	Output	1

6. Address map

See Figure 4. At the default geometry, `ADDR_MAP 0` gives `bank = {addr[24], addr[10]}`, `row = addr[23:11]`, `column = addr[9:0]`.

The Avalon address is a **word** address, matching the core this replaces, so a system's existing address assignments carry over unchanged.

7. Performance

Measured on `benchmark/`, identical stimulus, both controllers. ISSI IS42S16320D at 100 MHz, 16-bit bus, CAS 3, 4096 operations per pattern. Theoretical peak 200 MB/s.

Pattern	Intel's core	This core	Speedup
seq write	194.3	197.6	1.02×
seq read	194.0	196.0	1.01×
seq read/write	21.9	78.9	3.6×
same-row rd/wr	21.9	78.9	3.6×
bank+row walk	21.9	65.5	3.0×
4-bank same row	21.9	195.5	8.9×
random	22.0	44.4	2.0×

0 data errors, 0 timing violations, both controllers.

Sequential streaming was already finished and is essentially unchanged. The gain is entirely in mixed and scattered traffic.

Note

These are simulation figures. The first two rows reproduce the 194 MB/s the SDRAM example measured on hardware, which is the check that the harness measures the right thing — but no figure in this table has itself been observed on a board.

8. Verification

Flow	What it covers	Status
<code>simulation/verilator/run_sim.sh</code>	Lint, timing-checker self-test, testbench across 8 configurations, lint in 4 geometries	14 checks, 1152 testbench assertions, passing
<code>tb/avalon_mm_sdr_controller_tb.sv</code>	144 checks per configuration, asserting on the command stream as well as the data	Passing
<code>tb/avalon_mm_sdr_controller_sva.sv</code>	Avalon protocol, command legality, DQ contention, row bookkeeping	Passing
<code>tb/sdr_timing_check.sv</code>	tRC, tRAS, tRP, tRCD, tRRD, tWR, tMRD	Passing, with a 17-check threshold self-test
<code>benchmark/</code>	Throughput against the core being replaced	Passing
<code>example/de10_lite_rtl</code>	Board-level demonstration, 9 phases	58 checks passing in simulation
<code>simulation/questa/run_sim.tcl</code>	Coverage and assertion non-vacuity	Not run — no licence available
Hardware	Everything	Not run

8.1 The testbench checks the mechanism, not only the data

A controller that closed and reopened a row before every access would return correct data and pass any integrity check. So the testbench counts commands: four banks with one row open each, accessed in rotation, is asserted to cost exactly four ACTIVATES and zero PRECHARGES.

8.2 Proven by fault injection

Six faults were introduced into the controller and each was caught by the layer meant to catch it:

Fault	Caught by
Byte enables ignored	Directed test
Read/write turnaround removed	SVA
tRCD gate removed	Timing checker
ACTIVATE forgets its row	SVA
Refresh never forced under load	SVA
Read data captured a cycle late	Directed test

9. Limitations and known gaps

Limitation	Detail
Never run on hardware	No Quartus licence in the development environment. No f_MAX, no resource figures, no board result
No bursting	The slave is non-bursting; every transfer is one word. Bursts would amortise the read/write turnaround and are the main remaining performance work
No reordering	Transfers are served in order. A reorder buffer grouping same-direction accesses is the other half of that work
One chip select	Multi-device configurations are not supported
Single clock domain	No clock-crossing on the slave port
One device preset	Only the DE10-Lite's part. Others need datasheet figures
Questa flow unverified	Written to the same pattern as the firewall cores but never executed here

10. Revision history

Document version	Core version	Date	Change
1.0	1.0	August 2026	First release